

Figure 3: Responsibilities of an architect

Manager to get more business from the existing account (called account mining).

- Understands organisational sensitivities, and avoids rubbing people the wrong way, even if it is about selling an idea.
 - Helps in keeping the client satisfied on the technical front.
 - Serves as a single point of contact with the client on technical issues.
 - Understands the client's road map, and also the road map of the parent company. Identifies business opportunities.
- In a product development organisation, an ideal architect:
- Is more focused on a single technology area (deep-dive). For example, Data Link Layer Design, RTP (Real-time Transport Protocol), etc.
 - Will have in-depth expertise in the chosen areas, and is usually an authority in the chosen area, across the world.
 - Replies to the Request for Proposal (RFP) sent by the client, and contributes to specifications of relevant standards bodies such as 3rd Generation Partnership Project (3GPP2) and IEEE. Architects also contribute to open source operating systems in areas such as real-time behaviour, performance enhancement and others.
- The architect in a service organisation, too, performs these functions.

Responsibilities of an architect

In addition to the architects who are specialists in a particular area, some product and service organisations have product architects who will have all the five perspectives—solutions, network, systems, hardware and software—as explained in the previous article (Linux For You, April 2010 issue, Page 55).

All the crucial responsibilities of an architect [1, 2] are captured in Figure 3.

On top of the operational activities they perform, architects share their outstanding work with the community through paper presentations in reputed journals and conferences. This not only helps them gain recognition in the architect and technology community, but also serves the social cause of giving back to the system.

Architects are also expected to empathise with

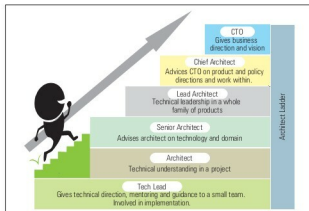


Figure 4: Career Path for an Architect

problems faced by the business, and usually are able to understand the business constraints. They are aware of the economies of scale, economics of product development, the business model and the revenue model. Without an understanding of these underlying ideas, it is very difficult to have the big picture around what they are creating.

Ultimately, in a business, the product/service that they are creating should create value for the customer. Only then will the organisation, of which the architect is a part, stand to gain. The most famous question thrown by the BDM (Business Development Manager) to the architect is, “Will this sell?” The architect’s equally famous response is, “You have to tell me.” But, instead, the paradigm should have been, let us work together to create value for the customer.

The authors also feel that in the process of working with the business, the architect should also not violate the fundamental principles of software and hardware development, and should correctly estimate the resources required to build the product.

The architects should encourage the writing of simple and clean code, and should understand their responsibilities. Expanding on the legendary computing science guru, Edsger Wybe Dijkstra’s ideas of beauty, we can say:

- An architect should create an inherently good design, so that there is beauty in the final structure.
- Architects should separate the concerns related to their technical output and business drivers.

The career path of an architect is transparent in most organisations. Let us look at this in greater detail.

Career path of an architect

Lead architects or chief architects guide the business on how to build the next product, and what it takes to do it. They also attend conferences to visualise what the next product on the horizon is. If you are a chief architect in a chosen area for long, and you understand the nuances of the business fairly well, then you can become the Chief Technology